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外文译文题目: 基于K均值聚类算法和减法聚类算法的图像分割

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Image Segmentation using K -means Clustering Algorithm and
Subtractive Clustering Algorithm

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基于 K 均值聚类算法和减法聚类算法的图像分割

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摘要：图像分割是将一幅图像分成不同的组。在基于聚类的图像分割领域已经进行了大量的研究。有不同的方法，其中最流行的方法是 K 均值聚类算法。K 均值聚类算法是一种无监督算法，用于将感兴趣区域从背景中分割出来。但在应用 K 均值算法之前，首先对图像进行部分拉伸增强，以改善图像的质量。减法聚类法是一种基于数据点的潜在值生成质心的数据聚类方法。所以使用减法聚类产生初始中心，这些中心用于 K 均值算法对图像进行分割。最后，对分割后的图像进行中间滤波，去除图像中不需要的区域。

关键词： 图像分割;K 均值聚类;中值滤波;局部对比度拉伸;减法聚类。

1.介绍

图像分割是面向决策应用中最常用的对图像像素进行正确分类的方法之一。它将一幅图像分割成若干离散的区域，使每个区域的像素具有高度的相似性，区域之间具有高度的对比度。它在医疗保健、图像处理、交通图像、模式识别等诸多领域都是一种有效的工具。图像分割有不同的技术，如基于阈值的图像分割、基于边缘的图像分割、基于聚类的图像分割、基于神经网络的图像分割。在不同的技术中，最有效的方法之一是聚类法。同样不同类型的聚类法：K 均值聚类，Fuzzy C-means 聚类、Mountain 聚类法以及减法聚类法。

K 均值聚类法最常用的聚类算法之一。它比分层聚类更简单，计算速度更快。它也适用于大量的变量。但是对于不同数量的聚类，它会产生不同的聚类结果。因此需要初始化适当的簇数 k 。同样，需要初始化 k 个质心。初始质心值的不同会导致不同的聚类。因此选择合适的初始质心也是一项重要的任务。

如今，如想分割已成为医学领域的重要工具之一，用于从背景中提取出感兴趣区域。因此，医学图像使用不同的技术进行分割，并且处理输出用于医学上的进一步分析。但是医学图像的原始形式是由计算机中的一组数字来表示的，这些数字表示相关物理量的值，这些物理量表示不同类型身体部位之间的对比。医学图像的处理和分析有助于将原始图像转换为可量化的符号形式，提取有意义的定性信息以辅助诊断，以及整合来自多种成像模式的互补数据。图像分割是医学分析中最基本的问题之一，它识别出图像中器官或异常区域等物体的边界。分割的结果使形状分析、体积变化检测和精确的放疗处理工厂成为可能。

2.相关工作

在图像分割领域，人们利用不同的方法已经做了很多工作。很多都是基于不同的应

用进行图像分割的。到目前为止在图像分割领域已经实现了很多方法，用不同的方法来初始化中心，K 均值聚类算法是最简单的聚类算法之一。许多研究人员也在尝试开发出比现有方法更高效、分割效果更好的新方法。本文对一些现存的近期作品进行了讨论。

Pallavi Purohit 和 Ritesh Joshi 介绍了一种新的高效的 K 均值聚类算法。他们提出了一种通过减少最终集群的均方误差而不太大幅增加执行时间来生成集群中心的新的聚类中心生成方法，。它在不牺牲执行时间的前提下减小了均方误差。经过大量的比较，可以得出密集数据集比稀疏数据集的精度更高的结论。

Alan Jose, S. Ravi 和 M. Sambath 提出了使用 K 均值聚类和模糊 c 均值算法及其面积计算的脑肿瘤分割方法。他们将过程分为图像预处理、高级 k 均值和模糊 c 均值以及特征提取三个部分。首先，利用滤波器实现预处理，提高图像质量，然后使用提出的改进 K 均值算法，模糊 c 均值算法对图像进行聚类。再将得到的分割图像用于感兴趣区域的特征提取。他们使用 MRI 图像进行分析，并计算图像中提取的肿瘤区域的大小。

Madhu Yedla, Srinivasa Rao Pathakota, t.m. Srinivasa 提出了改进初始中心的增强 K-means 聚类算法。提出了一种寻找初始质心的新方法，它提供了一种有效的将数据点分配到合适簇的方法，同时降低时间复杂度。他们证明了他们提出的算法比原来的 K 均值聚类算法具有更高的精度和更少的计算时间。该算法不需要任何额外的输入，如阈值。但该算法仍然需要初始化簇数 k，并建议确定 k 的取值作为今后的工作之一。

K.A.Abdul Nazeer 和 M.P.Sebastian 提出了一种增强算法来提高 K 均值聚类算法的精度和效率。他们提出了一种改进的 K 均值算法，它结合了由两种方法组成的系统方法。第一个任务是找到初始质心，另一个是将数据点分配给簇。他们取了不同的初始质心测试了执行时间和精度。结果表明，该算法在不牺牲聚类精度的前提下降低了时间复杂度。

3. 使用部分对比度拉伸增强对比度

用于分析的医学图像可能有其自身的弱点，如模糊或低对比度。为了提高图像的质量和对比度，采用了局部空间上浆(PCS)等对比度增强技术。它是通过拉伸和压缩过程来完成的。应用该技术，将下阈值和上阈值的像素范围映射到一个新的像素范围，并线性拉伸到新的下阈值范围内的一个较宽的像素范围，其余像素进行压缩(图 1)。

4.减法聚类算法

减法聚类是一种根据周围数据点的密度找到最优数据点来定义聚类质心的方法。该方法是由 Chiu 提出的 Mountain 方法的延伸。Mountain 法非常简单有效，它估计集群

中心的数量和初始位置。它将数据空间分布到网格点上,并根据每个数据点与实际数据点的距离计算每个数据点的潜力。因此,附近数据点较多的网格点具有较高的潜在价值。因此,选择潜在值最高的网格点作为第一个聚类中心。因此,在选择第一个簇中心后,我们将通过计算剩余格点中的最高潜在值来寻找第二个簇中心。由于靠近第一个集群中心的网格点会降低其潜在价值,所以下一个集群中心将会是在第一个集群中心网格点附近有多个数据点的网格。因此,获取新的聚类中心并降低周围网格点电位的过程重复进行,直到所有网格点电位都降到阈值以下。因此,该方法是寻找聚类中心最简单、最有效的方法之一。但随着数据维数的增加,其计算复杂度呈指数增长。因此,减法聚类算法解决了与山法相关的计算方法。该方法以数据点作为聚类中心的候选点,计算量与问题规模成正比。

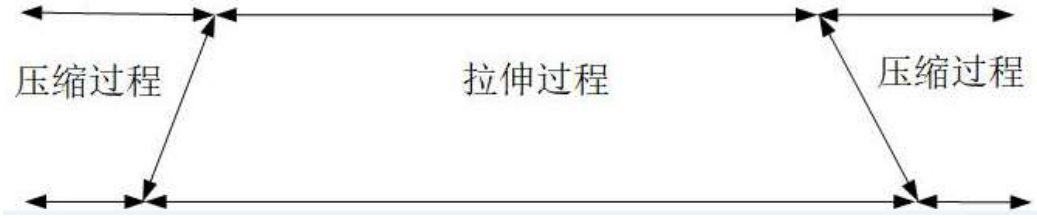


图 1. 部分对比拉伸工艺

假设有一个 n 个数据点的集合: $X = \{x_1, x_2, x_3 \dots x_n\}$ 。然后将每个点视为一个潜在的聚类中心。定义潜在的数据点的 x_n 为:

$$P_n = \sum_{i=1}^n e^{\frac{-4x_n - x_i^2}{r_a^2}} \quad (1)$$

其中 r_a 是数据空间中的超球簇半径,它是一个正常数,用来定义邻域。 $\|\cdot\|$ 为欧氏距离。所以数据点的势的度量就是到所有其他数据点的距离函数的度量。

求出每个数据点的势后,选择势最大的数据点作为第一个聚类中心。假设 x_1 和 p_1 分别为第一聚类中心及其对应的势。然后根据下式修正各数据点的势。

$$P_n = P_n - P_1 e^{\frac{-4x_n - x_1^2}{r_b^2}} \quad (2)$$

r_b 是数据空间中的超球补偿半径,它是一个正常数。在此每个数据点都减去一个势,作为到第一个聚类中心的距离的函数。所以靠近第一个聚类中心的数据点,其潜力会大大降低,因此选择为下一个聚类中心的机会也会减少。计算每个数据点的修正势后,找到下一个最高势作为下一个聚类中心。因此,这些过程将继续,直到获得足够数量的集群中心。

5.K 均值聚类算法

聚类是一种将一组数据划分为特定数量组的方法。最常用的方法是 K 均值聚类。在 K 均值聚类中，它将一个数据集合划分为一个 k 个数的数据组。它将给定的数据集分类为 k 个不相交的聚类。K 均值聚类算法由两个独立的阶段组成。在第一阶段，它计算 k 个质心，在第二阶段，它把每个点带到离各自数据点最近的质心的簇中。有不同的方法来定义最近的质心的距离，其中最常用的方法是欧氏距离。分组完成后，会重新计算每个新质心聚类并基于该质心，计算每个中心与每个数据点之间的新的欧氏距离，并分配聚类中欧氏距离最小的点。分区中的每个集群都由其成员对象和它的质心定义。每个簇的质心是到该簇中所有物体的距离之和最小的点。K 均值是一种迭代算法它最小化所有簇中每个对象到簇心的距离之和。

让我们考虑一幅分辨率为 $x \times y$ 的图像，该图像必须聚类成 k 个簇。设 $p(x, y)$ 为待聚类的输入像素， c_k 为聚类中心。K 均值聚类算法如下：

1.初始化簇 k 和中心的个数。

2.对于图像中的每个像素，使用下面给出的关系计算图像中心与每个像素之间的欧氏距离 d。

$$d = \|p(x, y) - c_k\| \quad (3)$$

3.根据距离 d 将所有像素分配到最近的中心

4.在所有像素被分配后，使用下面给出的关系重新计算中心的新位置。

$$c_k = \frac{1}{k} \sum_{y \in c_k} \sum_{x \in c_k} p(x, y) \quad (4)$$

5.重复此过程，直到满足公差或误差值为止。

6.将簇像素重新塑造成图像。

尽管 K 均值具有易于实现的巨大优势，但它也有一些缺点。最终聚类结果的质量取决于初始质心的选择。所以如果初始质心是随机选取的，对于不同的初始质心会得到不同的结果。所以我们会仔细选择初始的中心，这样我们就能得到我们想要的分割。计算复杂度是我们在设计 K 均值聚类时需要考虑的另一个术语。它依赖于数据元素的数量、集群的数量和迭代的数量。

6.中值滤波

中值滤波用于去除噪声，以获得无噪声图像。分割完成后，分割后的图像可能仍然会出现一些不需要的区域或噪声。因此，为了使分割后的图像质量更好，需要对分割后

的图像进行中值滤波。我们可以使用 $n \times n$ 的不同邻域，但通常使用 $n=7$ 的邻域，因为大的邻域会产生更严重的平滑。

7.采取的算法

所提出的算法由部分对比度拉伸、减法聚类、K 均值聚类和均值滤波组成。用于分割的医学图像大多对比度较低。因此对比度拉伸被用来提高图像的质量。在改进图像质量后，根据图像的潜在价值，采用减法聚类算法生成中心。中心的数量是根据聚类的数量 k 来生成的。这个中心在 K 均值算法中被用作初始中心。利用 K 均值算法，将图像分割成 k 个簇。图像分割后，图像中仍然会包含一些不需要的区域或噪声。利用均值滤波去除这些噪声。本文算法如下图(图 2)所示。

- 1.加载要分割的图像。
- 2.应用局部对比拉伸。初始化簇数 k 。
- 3.利用公式(1)计算图像中每个像素值的势。
- 4.求步骤 3 中的最大势，设该点为第一中心聚类，其对应的势为最大势。
- 5.利用公式(2)基于第一个聚类中心更新其他剩余像素的潜在值。

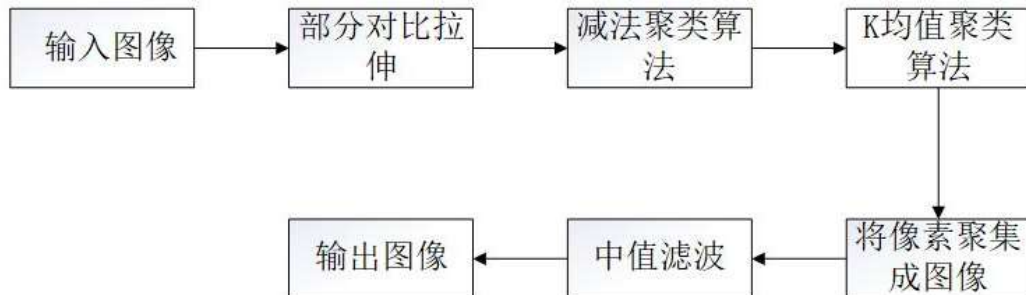


图 2.建议的算法框图

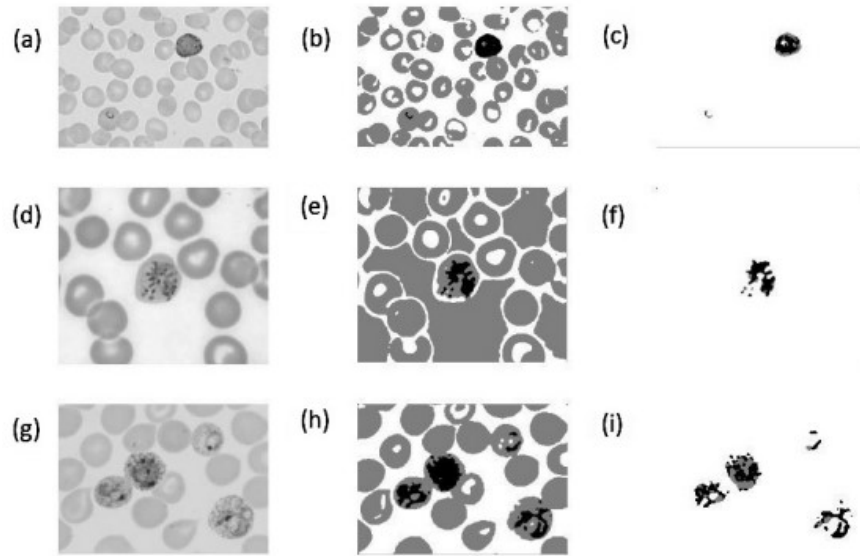


图 3.(a), (d), (g)原始图像;(b), (e), (h)K 均值算法;(c), (f), (i)建议的算法

6.再次求出第 4 步中的最大势，并将该点设为第二点。

7.继续这个过程，直到找到 k 个簇。

8.在 K 均值聚类算法中使用 k 中心作为初始中心。

9.利用关系式(3)求出图像中每个质心到每个像素的欧氏距离。

10.将相对于质心的距离最小的像素分配到你相应的质心簇。

11.根据公式(4)重新计算新的中心位置。

12.重复步骤 10-12，直到满足公差或误差值。

13.将集群重塑为映像。

14.对分割后的图像进行中值滤波，去除不需要的噪声或区域。

8.实验结果

我们使用不同类型的医学图像进行分析。在医学图像分析中，主要是从背景中分割出感染区域或感兴趣区域。我们用感染的血细胞来分析就像疟疾感染的血细胞一样。利用 Matlab 实现了该算法。我们将 K 均值算法与本文算法的结果进行比较，如图 3 所示。

在减法算法中，我们可以找到簇的数量 k 。但是我们没有找到簇的数量，而是根据图像的类型给出了用户定义的簇的数量 k 。这里我们取簇数， $k=3$ 。现在在减法簇中，我们可以使用不同的超球簇半径 r_a 和超球惩罚半径值 r_b 来调整输出图像。我们知道 r_a 定义了邻域距离，所以改变这个常数会影响输出。在图 4 中，我们展示了不同 r_a 值的输出图像。

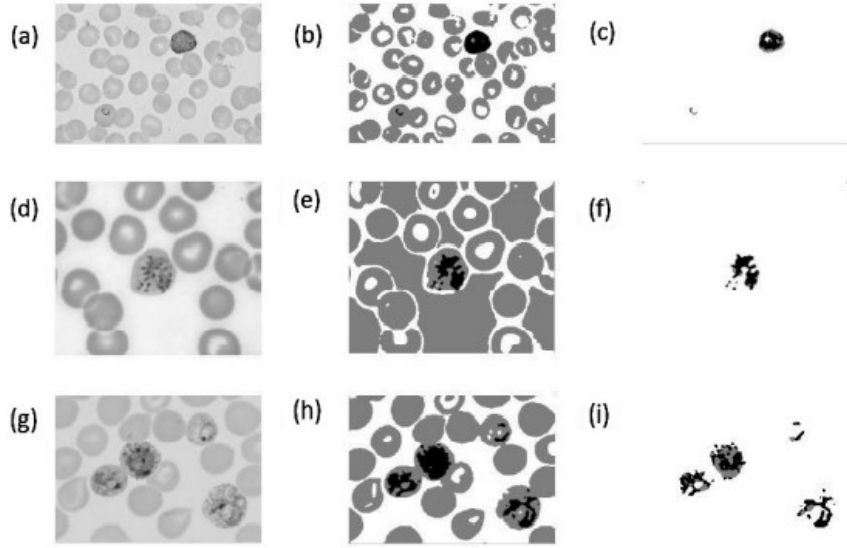


图 4.(a), (c), (i)原始图像(b), (f), (j); $r_b = 0.1$ 时的建议算法;(c),(g),(k) $r_b = 0.5$ 时的建议算法;(d),(h),(l) $r_b = 1.2$ 时的建议算法

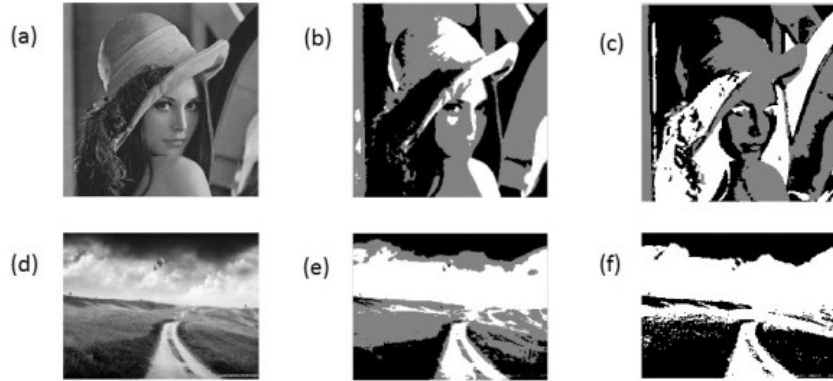


图 5.(a),(d)原始图像;(b),(e)K 均值算法;(c),(f)建议的算法

我们取了不同的 r 值, 根据 r 的值, 从图中可以观察到不同的结果。所以我们在取 r_a 值的时候要非常小心。我们还可以通过改变 r_a 的值来调整输出结果。同样地, 我们也可以通过改变球体惩罚半径 r_b 的值来调整输出。最后我们在其他一些图像中检查了结果, 分割后的输出结果如图 5 所示。从图中我们可以看出, 相比经典的 K 均值算法, 本文算法的输出结果具有更好的分割结果。利用均方根误差(RMSE)和峰值信噪比(PSNR)测量值分析分割后图像的质量。

1. 均方根误差: 它已被用作输出图像的标准性能测量。它给出了输出图像与输入图像的偏差

$$RMSE = \sqrt{\frac{1}{n_x n_y} \frac{\sum_{x=0}^{n_x-1} \sum_{y=0}^{n_y-1} [(r(x, y)) - t(x, y)]^2}{\sum_{x=0}^{n_x-1} \sum_{y=0}^{n_y-1} [r(x, y) - t(x, y)]^2}} \quad (5)$$

图像	根均方误差 (建议算法)	峰信噪比 (建议算法)	根均方误差 (K 均值算法)	峰信噪比 (K 均值算法)
血液样本 1	0.0077	34.80	0.0081	34.63
血液样本 2	0.0084	34.79	0.0144	32.44
血液样本 3	0.0073	34.43	0.0079	33.99
莉娜样本	0.0017	35.77	0.0041	31.9
视觉样本	0.0011	38.235	0.0036	32.94

2.峰值信噪比：峰值信噪比是最大可获得功率与影响图像相似性的破坏噪声之间的比例。它用于测量输出图像的质量。

$$PSNR = 10 \cdot \log_{10} \left[\frac{\max(r(x, y))^2}{\frac{1}{n_x n_y} \sum_0^{n_x-1} \sum_0^{n_y-1} [(r(x, y)) - t(x, y)]^2} \right] \quad (6)$$

其中 $r(x, y)$ 为输入图像， $t(x, y)$ 为分割后的图像。RMSE 值越小，图像质量越好，PSNR 值越小，图像质量越差。分割后图像的 RMSE 和 PSNR 值如表 1 所示。

计算了经典 K 均值算法的均方根误差(RMSE)和峰值信噪比(PSNR)。RMSE 值变得很低，PSNR 值达到 30 以上，当两种方法进行比较时，发现所提出的方法具有更好的效果。由此可见，该算法得到的输出图像质量较好。

9.结论

我们采用 K 均值聚类算法对图像进行分割，利用减法聚类生成初始质心。同时利用部分对比度拉伸来改善原始图像的质量，利用中值滤波来改善分割后的图像。并将最终分割结果与 K 均值聚类算法进行比较，可以看出本文提出的聚类算法具有较好的分割效果。输出图像也可以通过改变超球簇半径来调整，我们可以得出结论，通过改变超球簇半径，我们可以得到不同的输出。所以我们应该非常仔细地考虑超球团的值。最后对 RMSE 和 PSNR 进行了检验，发现它们分别具有较小和较大的值，这是图像分割质量良好的条件。并将本文方法与经典 K 均值聚类算法与 RMSE 和 PSNR 进行比较，发现本文方法具有更好的性能结果。未来，我们可以通过形态学运算进一步提高输出图像的质量，得到更好的性能度量。我们还可以使用减法聚类算法实现不同的聚类方法。最后对不同领域的图像分割进行实现和分析。

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Image Segmentation using K -means Clustering Algorithm and Subtractive Clustering Algorithm

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Abstract

Image segmentation is the classification of an image into different groups. Many researches have been done in the area of image segmentation using clustering. There are different methods and one of the most popular methods is k -means clustering algorithm. K -means clustering algorithm is an unsupervised algorithm and it is used to segment the interest area from the background. But before applying K -means algorithm, first partial stretching enhancement is applied to the image to improve the quality of the image. Subtractive clustering method is data clustering method where it generates the centroid based on the potential value of the data points. So subtractive cluster is used to generate the initial centers and these centers are used in k -means algorithm for the segmentation of image. Then finally medial filter is applied to the segmented image to remove any unwanted region from the image.

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Keywords: Image segmentation; K -means clustering; Median filter; Partial contrast stretching; Subtractive clustering.

1. Introduction

Image segmentation is one of the mostly used methods to classify the pixels of an image correctly in a decision oriented application. It divides an image into a number of discrete regions such that the pixels have high similarity in each region and high contrast between regions. It is a valuable tool in many field including health care, image processing, traffic image, pattern recognition etc. There are different techniques for image segmentation like threshold based, edge based, cluster based, neural network based¹. From the different technique one of the most efficient methods is the clustering method. Again there are different types of clustering: K -means clustering, Fuzzy C -means clustering, mountain clustering method and subtractive clustering method.

One of most used clustering algorithm is k -means clustering. It is simple and computationally faster than the hierarchical clustering. And it can also work for large number of variable. But it produces different cluster result for different number of number of cluster. So it is required to initialize the proper number of number of cluster, k^2 . Again, it is required to initialize the k number of centroid. Different value of initial centroid would result different cluster. So selection of proper initial centroid is also an important task.

Nowadays image segmentation becomes one of important tool in medical area where it is used to extract or region of interest from the background. So medical images are segmented using different technique and process outputs are

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used for the further analysis in medical. But medical images in their raw form are represented by the arrays of numbers in the computer³, with the number indicating the values of relevant physical quantities that show contrast between different types of body parts. Processing and analysis of medical images are useful in transforming raw images into a quantifiable symbolic form, in extracting meaningful qualitative information to aid diagnosis and in integrating complementary data from multiple imaging modalities. And one of the fundamental problems in medical analysis is the image segmentation which identifies the boundaries of objects such as organs or abnormal region in images. Results from the segmentation make it possible for shape analysis, detecting volume change, and making a precise radiation therapy treatment plant.

2. Related Work

There have been many works done in the area of image segmentation by using different methods. And many are done based on different application of image segmentation. K -means algorithm is the one of the simplest clustering algorithm and there are many methods implemented so far with different method to initialize the centre. And many researchers are also trying to produce new methods which are more efficient than the existing methods, and shows better segmented result. Some of the existing recent works are discussed here.

Pallavi Purohit and Ritesh Joshi⁴ introduced a new efficient approach towards K -means clustering algorithm. They proposed a new method for generating the cluster center by reducing the mean square error of the final cluster without large increment in the execution time. It reduced the means square error without sacrificing the execution time. Many comparisons have been done and it can conclude that accuracy is more for dense dataset rather than sparse dataset.

Alan Jose, S. Ravi and M. Sambath⁵ proposed Brain Tumor Segmentation using K -means Clustering and Fuzzy C -means Algorithm and its area calculation. In the paper, they divide the process into three parts, pre-processing of the image, advanced k -means and fuzzy c -means and lastly the feature extraction. First pre-processing is implemented by using the filter where it improves the quality of the image. Then the proposed advance K -means algorithm is used, followed by Fuzzy c -means to cluster the image. Then the resulted segment image is used for the feature extraction for the region of interest. They used MRI image for the analysis and calculate the size of the extracted tumor region in the image.

Madhu Yedla, Srinivasa Rao Pathakota, T. M. Srinivasa⁶ proposed Enhancing K -means clustering algorithm with improved initial center. A new method for finding the initial centroid is introduced and it provides an effective way of assigning the data points to suitable clusters with reduced time complexity. They proved their proposed algorithm has more accuracy with less computational time comparatively original k -means clustering algorithm. This algorithm does not require any additional input like threshold value. But this algorithm still initializes the number of cluster k and suggested determination of value of k as one of the future work.

K. A. Abdul Nazeer, M. P. Sebastian⁷ proposed an enhanced algorithm to improve the accuracy and efficiency of the k -means clustering algorithm. They present an enhanced k -means algorithm which combines a systematic method consisting two approaches. First one is finding the initial centroid and another is assigning the data point to the clusters. They have taken different initial centroid and tested execution time and accuracy. From the result it can be conclude that the proposed algorithm reduced the time complexity without sacrificing the accuracy of clusters.

3. Contrast Enhancement using Partial Contrast Stretching

Medical images which have been used for the analysis may have their own weakness such as blurred or low contrast. So a contrast enhancement technique such as Partial Spatial Starching (PCS) is used to improve the image quality and contrast of the image⁸. It is done by stretching and compression process. By applying this technique, the pixel range of lower threshold value and upper threshold value will be mapped to a new pixel range and stretched linearly to a wide range of pixels within new lower stretching value, and the remaining pixels will experience compression (Fig. 1).

4. Subtractive Clustering Algorithm

Subtractive clustering is a method to find the optimal data point to define a cluster centroid based on the density of surrounding data points⁹. This method is the extension of Mountain method, proposed by Chiu¹⁰. Mountain method is

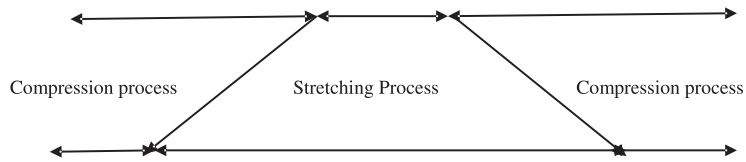


Fig. 1. Partial contrast stretching process.

very simple and effective. It estimates the number and initial location of the cluster centers. It distribute the data space into gridding point and compute the potential for each data point base on its distance to the actual data point. So the grid point with many data point nearby will have high potential value. And so this grid point with highest potential value will be choose as first cluster centre. So after selecting the first cluster centre we will try to find the second cluster centre by calculating the highest potential value in the remaining grid points. As grid points near the first cluster center will reduce its potential value, the next cluster center will be grid with many data point nearby other than first cluster center grid point. So this procedure of acquiring new cluster center and reducing the potential of surrounding grid point repeat until potential of all grid points falls below a threshold value. So this method is one of the simplest and effective methods to find the cluster centers. But with increase in the dimension of data, its computation complexity grows exponentially. So, subtractive clustering algorithm solves the computational method associated with mountain method. It uses data points as the candidates for cluster centre and the computation of this method is proportional to the problem size.

Consider a collection of n data points: $X = \{x_1, x_2, x_3 \dots x_n\}$. Then each point is considered as a potential cluster center. The potential of data point's x_n is defined as:

$$P_n = \sum_{i=1}^n e^{\frac{-4x_n - x_i^2}{r_a^2}} \quad (1)$$

where r_a is hyper sphere cluster radius in data space and it is a positive constant which is used to define the neighbourhood. The symbol $\|\cdot\|$ denotes the Euclidean distance. So the measure of the potential for the data point is a measure of function of distance to all other data points.

After finding the potential of each data points, select the data point with maximum potential as the first cluster centre. Let us consider x_1 and p_1 as first cluster centre and its corresponding potential respectively. Then revise the potential of each data point by using the formula given below.

$$P_n = P_n - P_1 e^{\frac{-4x_n - x_1^2}{r_b^2}} \quad (2)$$

r_b is the hyper sphere penalty radius in data space and it is a positive constant. Here an amount of potential is subtracted from each data point as a function of distance from the first cluster center. So the data points near the first cluster center will have greatly reduced potential, and therefore it have less chance to select as next cluster center. After calculating the revise potential of each data points, find the next highest potential as the next cluster center. So these processes continue until a sufficient number of cluster centre are obtained.

5. K-Means Clustering Algorithm

Clustering is a method to divide a set of data into a specific number of groups. It's one of the popular method is k -means clustering. In k -means clustering, it partitions a collection of data into a k number group of data^{11,12}. It classifies a given set of data into k number of disjoint cluster. K -means algorithm consists of two separate phases. In the first phase it calculates the k centroid and in the second phase it takes each point to the cluster which has nearest centroid from the respective data point. There are different methods to define the distance of the nearest centroid and one of the most used methods is Euclidean distance. Once the grouping is done it recalculate the new centroid of each

cluster and based on that centroid, a new Euclidean distance is calculated between each center and each data point and assigns the points in the cluster which have minimum Euclidean distance. Each cluster in the partition is defined by its member objects and by its centroid. The centroid for each cluster is the point to which the sum of distances from all the objects in that cluster is minimized. So K -means is an iterative algorithm in which it minimizes the sum of distances from each object to its cluster centroid, over all clusters.

Let us consider an image with resolution of $x \times y$ and the image has to be cluster into k number of cluster. Let $p(x, y)$ be an input pixels to be cluster and c_k be the cluster centers. The algorithm for k -means¹³ clustering is following as:

1. Initialize number of cluster k and centre.
2. For each pixel of an image, calculate the Euclidean distance d , between the center and each pixel of an image using the relation given below.

$$d = \|p(x, y) - c_k\| \quad (3)$$

3. Assign all the pixels to the nearest centre based on distance d .
4. After all pixels have been assigned, recalculate new position of the centre using the relation given below.

$$c_k = \frac{1}{k} \sum_{y \in c_k} \sum_{x \in c_k} p(x, y) \quad (4)$$

5. Repeat the process until it satisfies the tolerance or error value.
6. Reshape the cluster pixels into image.

Although k -means has the great advantage of being easy to implement, it has some drawbacks. The quality of the final clustering results is depends on the arbitrary selection of initial centroid. So if the initial centroid is randomly chosen, it will get different result for different initial centers. So the initial center will be carefully chosen so that we get our desire segmentation. And also computational complexity is another term which we need to consider while designing the K -means clustering. It relies on the number of data elements, number of clusters and number of iteration.

6. Median Filter

Median filtering is used as a noise removal in order to obtain a noise free image. After segmentation is done, the segmented image may still present some unwanted regions or noise. So to make the image a good and better quality, the median filter is applied to the segmented image. We can use different neighborhood of $n \times n$. But generally neighborhood of $n = 7$ is used because large neighborhoods produce more severe smoothing.

7. Proposed Algorithm

The proposed algorithm consists of partial contrast stretching, subtractive clustering, k -means clustering and median filter. Mostly the medical images which are used for segmentation have low contrast. So contrast stretching is used to improve the quality of the image. After improving the quality of image, subtractive clustering algorithm is used to generate the centers, based on the potential value of the image. Number of centre is generated based on number of cluster k . This centre is used as initial centre in k -means algorithm. Using the k -means algorithm, the image is segmented into k number of cluster. After the segmentation of image, the image can still contain some unwanted region or noise. These noises are removed by using the median filter. The proposed algorithm is followed as below (Fig. 2).

1. Load the image to be segmented.
2. Apply partial contrast stretching.
Initialize number of cluster, k .
3. Use equation (1) to calculate the potential for every pixel value of the image.
4. Find maximum potential in step 3 and set that point be first center cluster and its corresponding potential as maximum potential.
5. Use equation (2) to update the potential value of other remaining pixels based on the first cluster center.

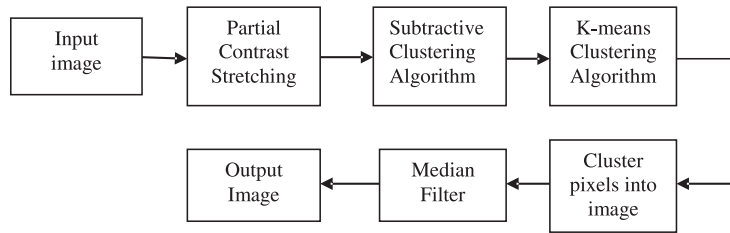


Fig. 2. Block diagram of the proposed algorithm.

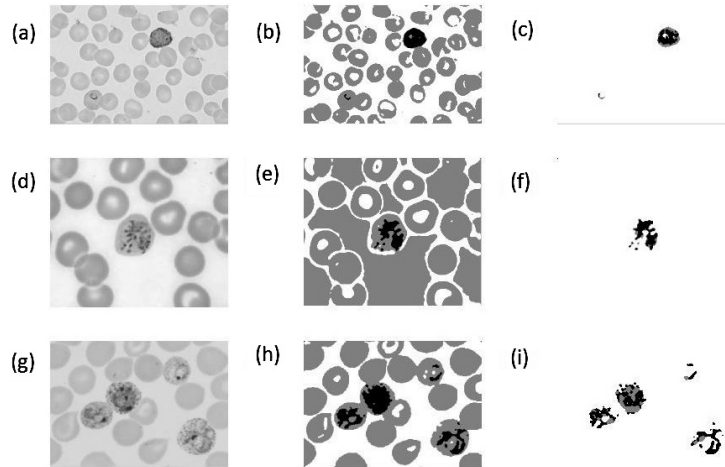


Fig. 3. (a), (d), (g) Original image; (b), (e), (h) *K*-means algorithm; (c), (f), (i) Proposed algorithm.

6. Again find the maximum potential in the step 4 and let that point be second point.
7. Continue the process until it finds the k number of cluster.
8. Used k centre as initial centre in the k -means clustering algorithm.
9. Find the Euclidean distance of each centroid from every pixel of the image using the relation (3).
10. Assign the pixel with minimum distance with respect to centroid to its respective cluster of the centroid.
11. Recalculate the new center location by using the equation (4).
12. Repeat the steps 10–12, until it satisfies the tolerance or error value.
13. Reshape the cluster into image.
14. Median filter is applied to the segmented image to remove any unwanted noise or region.

8. Results

We used different types of medical images for the analysis. In medical image analysis, mostly the infected areas or area of interest are segmented from background. We used infected blood cell like malaria infected blood cell for the analysis. Matlab is used to implement the proposed algorithm. We compare the result of k -means algorithm with proposed algorithm and it is shown in the Fig. 3.

In the subtractive algorithm we can find the number of cluster k . But instead of finding the number of cluster we have given user defined number of cluster k based on the type of the image. Here we have taken the number of cluster, $k = 3$. Now in the subtractive cluster we can tune the output image by using the different value of hyper sphere cluster radius, r_a and hyper sphere penalty radius, r_b . We know that r_a defines the neighborhood distance, so changing this constant will affect the output. In the Fig. 4 we have shown output images for different value of r_a .

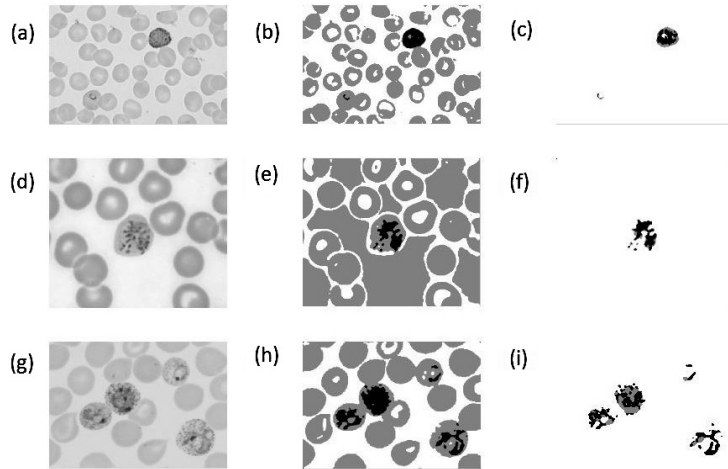


Fig. 4. (a), (e), (i) Original image; (b), (f), (j) Proposed algorithm with $r_b = 0.1$; (c), (g), (k) Proposed algorithm with $r_b = 0.5$; (d), (h), (l) Proposed algorithm with $r_b = 1.2$.

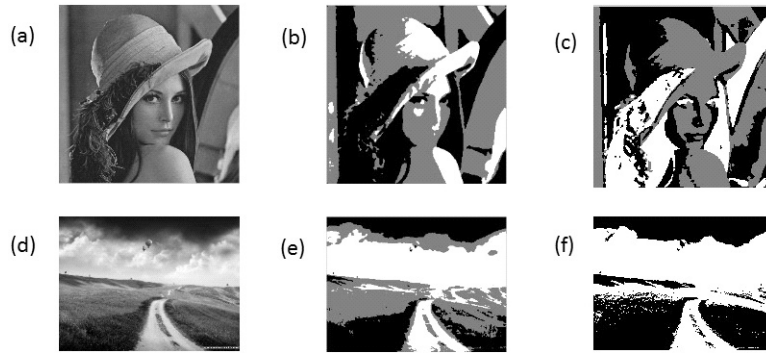


Fig. 5. (a), (d) Original image; (b), (e) K -means algorithm; (c), (f) Proposed algorithm.

We have taken different value of r and according to the value of r we get different result as observed from the figure. So we need to be very careful while taking the value of r_a . And we can also tune the output result by varying the value of r_a . In the same way we can also tune the output by varying the value of the sphere penalty radius, r_b .

And lastly we have check the result in some other images and the segmented output result are shown above in the Fig. 5. From the figure we observed that the output result using proposed algorithm has better segmentation result as compared to the classical k -means algorithm.

The quality of the segmented image is analyzed using the measurement value of Root Mean Square Error (RMSE) and Peak to Signal Noise Ration (PSNR)¹⁴.

1. Root Mean Square Error: It has been used as a standard performance measurement of the output image. It gives how much output image is deviated from the input image.

$$RMSE = \sqrt{\frac{1}{n_x n_y} \frac{\sum_{x=0}^{n_x-1} \sum_{y=0}^{n_y-1} [r(x, y)]^2}{\sum_{x=0}^{n_x-1} \sum_{y=0}^{n_y-1} [r(x, y) - t(x, y)]^2}} \quad (5)$$

Table 1. RMSE and PSNR values.

Image	RMSE (proposed algorithm)	PSNR (proposed algorithm)	RMSE (<i>K</i> -means algorithm)	PSNR (<i>K</i> -means algorithm)
First Blood Sample	0.0077	34.80	0.0081	34.63
Second Blood Sample	0.0084	34.79	0.0144	32.44
Third Blood Sample	0.0073	34.43	0.0079	33.99
Lena Sample	0.0017	35.77	0.0041	31.9
Scenery Sample	0.0011	38.235	0.0036	32.94

2. Peak to Signal Noise Ratio: The peak to signal noise ratio is the proportion between maximum attainable powers and the corrupting noise that influence likeness of image. It is used to measure the quality of the output image.

$$PSNR = 10 \cdot \log_{10} \left[\frac{\max(r(x, y))^2}{\sqrt{\frac{1}{n_x n_y} \frac{\sum_{x=0}^{n_x-1} \sum_{y=0}^{n_y-1} [(r(x, y))]^2}{\sum_{x=0}^{n_x-1} \sum_{y=0}^{n_y-1} [r(x, y) - t(x, y)]^2}}} \right] \quad (6)$$

where $r(x, y)$ is the input image and $t(x, y)$ is the segmented image. And the smaller value of RMSE means the image is of good quality and smaller value of PSNR means the image of poor quality. The value of RMSE and PSNR of the segmented image is given below in the Table 1.

The RMSE and PSNR value are calculated for classical *K*-means Algorithm as well as proposed method. The values of RMSE are getting very low and the value of PSNR is getting above 30 and when both methods are compared it is found that the proposed method has better result. So we can conclude that the output image resulted from the proposed algorithm are of good quality.

9. Conclusion

We have segmented an image by using *k*-clustering algorithm, using subtractive cluster to generate the initial centroid. At the same time partial contrast stretching is used to improve the quality of original image and median filter is used to improve segmented image. And the final segmented result is compare with *k*-means clustering algorithm and we can conclude that the proposed clustering algorithm has better segmentation. The output images are also tune by varying the hyper sphere cluster radius and we can conclude from that result that by varying the hyper sphere cluster radius we can get different output. And so we should take the value of hyper sphere cluster very carefully. Finally RMSE and PSNR are checked and observed that they have small and large value respective, which are the condition for good image segmentation quality. And comparison for RMSE and PSNR are done for proposed method and classical *K*-means algorithm and it is found that the proposed method have better performance result. In the future, we can improve the quality of the output image more by using the morphological operation and get better performance measurement. We can also implement different clustering method using subtractive clustering algorithm. And lastly we can implement and analyze in different areas of image segmentation.

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